

EPGP Academic Scheduling Optimization using Linear Programming

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Introduction

This formulation represents the scheduling problem for the IIM EPGP Program, accommodating different sections/areas, courses, time slots, and faculty members. The objective is to create an optimal schedule that satisfies all hard constraints while minimizing violations of a configurable consecutive-period soft constraint.

1 Sets and Indices

- $b \in B$: Set of buckets. Buckets are either section identifiers (in Sections mode) or area short-names (in Areas mode).
- $c \in C$: Set of Courses.
- $t \in T_b$: Set of time slots available for bucket b .
- $f \in F$: Set of Faculty members.
- $d \in D$: Set of unique Calendar Dates in the quarter.
- $pk \in PK_b$: Set of **period keys** for bucket b — unique integers that identify each scheduling period. The encoding is determined by the configured period unit:

- `period_unit = "weeks"`: $pk = \text{iso_year} \times 100 + \text{iso_week}$ (e.g. Week 10 of 2026 $\rightarrow$ 202610)
- `period_unit = "days"`: $pk = \text{date.toordinal}()$ (unique integer per calendar day)

Two period keys pk and pk' are **consecutive** if they represent immediately adjacent periods — successive ISO weeks (with correct year-rollover handling) or successive calendar days. Note that *arithmetic* $pk + 1$ is **not** the same as the consecutive successor for week-encoded keys (e.g. 202652 $\rightarrow$ 202701, not 202653). We write $\text{succ}^j(pk_0)$ for the j -th consecutive successor of pk_0 .

- $g \in G$: Set of course conflict groups (used by optional constraint $H7$).
- $C_g \subseteq C$: Set of courses in conflict group g .
- $B_g \subseteq B$: Set of buckets to which conflict group g applies.

2 Parameters

- $R_{b,c}$: Required number of sessions for bucket b in course c .
- $\text{Faculty}(b, c)$: The faculty member assigned to teach course c in bucket b .
- $\text{Date}(b, t)$: The calendar date associated with slot t for bucket b .
- $\text{Period}(b, t)$: The period key pk associated with slot t for bucket b (see PK_b above).

- $\text{Boundary}(b, t)$: The boundary-group key of slot t for bucket b . Controls where the consecutive counter resets:
 - `reset_boundary = "month"`: $\text{Boundary}(s, t) = \text{year} \times 100 + \text{month}$
 - `reset_boundary = "none"`: $\text{Boundary}(s, t) = 0$ (no reset; sequence is continuous)
- $\text{Unavailable}(f, d, t_f)$: Set-valued parameter (or indicator function) defining faculty unavailability at slot level; 1 if faculty f is unavailable on date d and time-slot start t_f , 0 otherwise. Includes both:
 - legacy full-day unavailability (entire date blocked), and
 - slot-specific unavailability (only certain time windows blocked within a date).

(used by optional constraint H6)
- M_f : Maximum sessions faculty f may teach per calendar day.
- M : Maximum number of **consecutive periods** in which the same course may be taught to the same scope without incurring a penalty (user-configured; default $M = 2$). A window of $M + 1$ consecutive same-boundary periods that all have active sessions constitutes one violation.
- α : Penalty weight for the session spreading rule (user-configured; default $\alpha = 0.1$). Controls the priority of spreading relative to the consecutive rule in the objective. A smaller α keeps spreading as a secondary preference; a larger α enforces it more aggressively. Recommended range: 0.01–0.5.
- D : Total number of calendar days in the scheduling horizon, $D = (\text{end_date} - \text{start_date}).\text{days} + 1$.
- $\mathcal{E}_{b,c,i}$: The set of slot indices t for bucket b and course c whose calendar date falls in epoch i , where epoch i covers the day-offset range $[i \cdot D/R_{b,c}, [(i+1) \cdot D/R_{b,c}] - 1]$ and $i \in \{0, 1, \dots, R_{b,c} - 1\}$. Only slots that are not excluded by faculty unavailability or area exclusion are included. An epoch with $\mathcal{E}_{b,c,i} = \emptyset$ carries no penalty and no variable is created for it.
- Scope: The consecutive-rule and auxiliary variables operate over a configurable *scope*:
 - Sections mode: $\text{scope} = (b, c)$ (bucket-course pair, i.e., per-section behavior)
 - Areas mode: $\text{scope} = c$ (course-level scope aggregated across buckets/areas)

3 Decision Variables

- $x_{b,c,t} \in \{0, 1\}$: 1 if bucket b is assigned course c at time slot t ; 0 otherwise. (In Sections mode this enforces one course per (b, t) ; in Areas mode multiple concurrent $x_{b,c,t} = 1$ for the same (b, t) are allowed unless forbidden by other constraints.)
- $y_{\text{scope}, pk} \in \{0, 1\}$: Auxiliary variable; 1 if the given *scope* has at least one session of the relevant course in period pk ; 0 otherwise.
- $p_{\text{scope}, pk_0} \in \{0, 1\}$: Penalty variable; 1 if the scope is active in *every* period of the window $(pk_0, \text{succ}(pk_0), \dots, \text{succ}^M(pk_0))$; 0 otherwise. Created only when the consecutive rule is enabled.
- $e_{b,c,i} \in \{0, 1\}$: Epoch-presence variable; 1 if at least one session of course c in bucket b falls within epoch i ; 0 otherwise. Created only when the spreading rule is enabled and $R_{b,c} \geq 2$, and only for epochs where $\mathcal{E}_{b,c,i} \neq \emptyset$.

4 Objective Function

The objective depends on which soft constraints are enabled in the configuration. Let $\mathbf{1}[\cdot]$ denote indicator conditions on the enabled flags.

General form:

$$\text{Minimize } Z = \underbrace{\sum_{\text{scope}} \sum_{pk_0} p_{\text{scope}, pk_0}}_{\text{consecutive penalty}} + \underbrace{\alpha \sum_{b \in B} \sum_{c \in C} \sum_{i=0}^{R_{b,c}-1} (1 - e_{b,c,i})}_{\text{spreading penalty}} \quad (1)$$

Each term is included only when its corresponding rule is enabled:

Consecutive	Spreading	Objective
Disabled	Disabled	$Z = 0$ (pure feasibility)
Enabled	Disabled	$Z = \sum_{\text{scope}, pk_0} p_{\text{scope}, pk_0}$
Disabled	Enabled	$Z = \alpha \sum_{b,c,i} (1 - e_{b,c,i})$
Enabled	Enabled	$Z = \sum_{\text{scope}, pk_0} p_{\text{scope}, pk_0} + \alpha \sum_{b,c,i} (1 - e_{b,c,i})$

The two terms are additive and independent. Because α is kept small, the consecutive rule remains dominant when both are active. The spreading term $(1 - e_{b,c,i}) = 1$ when epoch i is empty (a gap) and 0 when it contains at least one session. Minimizing the sum is therefore equivalent to filling as many epochs as possible, pushing the solver toward uniform distribution.

5 Hard Constraints

5.1 Fixed Hard Constraints (always applied)

- **H1: Session Requirement**

$$\sum_{t \in T_b} x_{b,c,t} = R_{b,c} \quad \forall b \in B, c \in C \quad (2)$$

Every bucket must complete the required number of sessions for every assigned course.

- **H2: At Most One Course per Slot**

$$\sum_{c \in C} x_{b,c,t} \leq 1 \quad \forall b \in B, t \in T_b \quad (\text{Sections mode only}) \quad (3)$$

In Sections mode each available time slot for a bucket (section) may be occupied by at most one course, so unused slots are allowed when total available capacity exceeds required sessions. This constraint is not applied in Areas mode, where areas may host parallel sessions in the same slot.

- **H3: No Faculty Cloning**

$$\sum_{\substack{(b,c): \\ \text{mathrmFaculty}(b,c)=f \\ \text{mathrmDate}(b,t)=d, \text{FromTime}(b,t)=\tau}} x_{b,c,t} \leq 1 \quad \forall f \in F, (d, \tau) \quad (4)$$

A faculty member cannot teach more than one bucket at the same date and start time.

- **H4: Maximum Daily Workload**

$$\sum_{b \in B} \sum_{c \in C} \sum_{\substack{t \in T_b: \\ \text{mathrmDate}(b,t)=d, \\ \text{mathrmFaculty}(b,c)=f}} x_{b,c,t} \leq M_f \quad \forall f \in F, d \in D \quad (5)$$

A faculty member may teach at most M_f sessions per calendar date across all buckets.

- **H5: Course Spacing (Daily)**

$$\sum_{\substack{t \in T_b: \\ \text{mathrmDate}(b,t)=d}} x_{b,c,t} \leq 1 \quad \forall b \in B, c \in C, d \in D \quad (6)$$

A bucket cannot be scheduled for the same course more than once on the same calendar day.

5.2 Optional Hard Constraints (can be enabled/disabled in configuration)

- **H6: Faculty Unavailability**

$$x_{b,c,t} = 0 \quad \text{if } (\text{Date}(b,t), \text{Time}(b,t)) \in \mathcal{U}_{\text{Faculty}(b,c)} \quad \forall b \in B, c \in C, t \in T_b \quad (7)$$

A class cannot be scheduled in any time slot (date,fromTime) marked as unavailable for the assigned faculty member. Full-day unavailability is modeled as all time slots of that date being included in $\mathcal{U}_f$.

- **H7: Course Conflict Groups (Bucket Scope)**

$$\sum_{b \in B_g} \sum_{c \in C_g} \sum_{\substack{t: \\ \text{Date}(b,t)=d, \\ \text{FromTime}(b,t)=\tau}} x_{b,c,t} \leq 1 \quad \forall g \in G, (d, \tau) \quad (8)$$

For each conflict group g , at most one course from C_g may run at any given date and start time across the applicable buckets B_g .

6 Soft Constraints (can be enabled/disabled in configuration)

6.1 Consecutive Sessions Rule

When the consecutive rule is enabled, we penalize every window in which the same course is taught to the same section in $M + 1$ or more consecutive same-boundary periods. The formulation uses auxiliary period-activity indicators y to link slot-level decisions x to the period-level penalty variables p .

Window Eligibility

A window $\mathcal{W} = (pk_0, \text{succ}(pk_0), \dots, \text{succ}^M(pk_0))$ of length $M + 1$ over (s, c) is **eligible** if and only if:

1. **Coverage:** All $M + 1$ period keys in $\mathcal{W}$ exist in PK_s for section s with valid slots mapped to course c .
2. **Consecutiveness:** Each successive pair in $\mathcal{W}$ are adjacent periods (next ISO week, or next calendar day), including correct year-boundary handling.
3. **Boundary confinement** (when `reset_boundary` $\neq$ "none"):

$$\text{Boundary}(s, t_0) = \text{Boundary}(s, t_j) \quad \forall j = 1, \dots, M \quad (9)$$

where t_0, t_j are representative slots in pk_0 and $\text{succ}^j(pk_0)$ respectively. This ensures the consecutive counter resets at month (or other configured) boundaries, so violations cannot span across them.

Auxiliary Constraints (linking y to x)

- **A1: Period Activity — Lower Bound**

$$y_{\text{scope},pk} \geq x_{b,c,t} \quad \forall b \in B, c \in C, t \in T_b \quad \text{where } \text{Period}(b,t) = pk \text{ and scope applies} \quad (10)$$

$y_{\text{scope},pk}$ is forced to 1 if *any* session within the scope is scheduled in period pk .

- **A2: Period Activity — Upper Bound**

$$y_{\text{scope},pk} \leq \sum_{\substack{t \in T_b \\ \text{Period}(b,t)=pk \\ \text{and scope applies}}} x_{b,c,t} \quad \forall pk \in \bigcup_b PK_b \quad (11)$$

$y_{\text{scope},pk}$ is forced to 0 if *no* session within the scope is scheduled in period pk .

Penalty Activation

- **P: Consecutive Period Penalty**

For every eligible window $\mathcal{W} = (pk_0, \text{succ}^1(pk_0), \dots, \text{succ}^M(pk_0))$ over the chosen scope:

$$p_{\text{scope}, pk_0} \geq \sum_{pk \in \mathcal{W}} y_{\text{scope}, pk} - M \quad (12)$$

Activation logic:

- If every period in $\mathcal{W}$ is active ($y = 1$ for all $M + 1$ periods): $\sum_{pk \in \mathcal{W}} y_{\text{scope}, pk} = M + 1$, so RHS = 1, forcing $p_{\text{scope}, pk_0} \geq 1$ — the penalty fires.
- If at most M periods are active: $\sum_{pk \in \mathcal{W}} y_{\text{scope}, pk} \leq M$, so RHS ≤ 0 . Since $p \geq 0$ always holds, the constraint is slack and p remains 0.

Special cases recovered by this general formulation:

- $M = 1$, `reset_boundary = "none"`: reduces to the classic *Alternate-Week Rule* — the same course cannot be taught in two consecutive weeks within the configured scope.
- $M = 2$, `reset_boundary = "month"`: implements the *Two-Consecutive-Week Rule within a Month* — within the chosen scope the course may appear in at most 2 consecutive weeks per month; a third consecutive week in the same month triggers a penalty.

6.2 Session Spreading Rule

When the spreading rule is enabled, we penalize every epoch that receives no session, encouraging the solver to distribute sessions uniformly across the scheduling horizon. The formulation introduces epoch-presence variables e that link slot-level decisions x to the epoch-level spreading penalty.

Epoch Definition

For each (b, c) pair with $R_{b,c} \geq 2$, the scheduling horizon of D days is divided into $R_{b,c}$ equal-length epochs. Epoch i covers calendar days in the offset range:

$$\left[\left\lfloor \frac{i \cdot D}{R_{b,c}} \right\rfloor, \left\lfloor \frac{(i+1) \cdot D}{R_{b,c}} \right\rfloor - 1 \right] \quad i \in \{0, 1, \dots, R_{b,c} - 1\} \quad (13)$$

The epoch index for a slot t is therefore:

$$\text{epoch}(b, t) = \min \left(\left\lfloor \frac{(\text{Date}(b, t) - \text{start_date}).\text{days} \times R_{b,c}}{D} \right\rfloor, R_{b,c} - 1 \right) \quad (14)$$

Slots excluded by faculty unavailability or area exclusion are omitted from $\mathcal{E}_{b,c,i}$. An epoch whose $\mathcal{E}_{b,c,i} = \emptyset$ is structurally unschedulable and is excluded from the penalty — no variable is created and no penalty is counted.

Auxiliary Constraints (linking e to x)

- **E1: Epoch Presence — Lower Bound**

$$e_{b,c,i} \geq x_{b,c,t} \quad \forall t \in \mathcal{E}_{b,c,i} \quad (15)$$

$e_{b,c,i}$ is forced to 1 if *any* session is placed in epoch i .

- **E2: Epoch Presence — Upper Bound**

$$e_{b,c,i} \leq \sum_{t \in \mathcal{E}_{b,c,i}} x_{b,c,t} \quad \forall b \in B, c \in C, i \in \{0, \dots, R_{b,c} - 1\} \quad (16)$$

$e_{b,c,i}$ is forced to 0 if *no* session falls in epoch i . This upper bound is critical: without it the solver could set $e_{b,c,i} = 1$ freely for all epochs, gaining reward without actually spreading sessions.

Penalty Contribution

- **S: Empty-Epoch Penalty**

For each schedulable epoch i of pair (b, c) :

$$\text{gap}_{b,c,i} = 1 - e_{b,c,i} \quad (17)$$

$\text{gap}_{b,c,i} = 1$ when epoch i is empty; 0 when it contains at least one session. The total spreading penalty contributed to the objective is:

$$\alpha \sum_{b \in B} \sum_{c \in C} \sum_{i=0}^{R_{b,c}-1} (1 - e_{b,c,i}) \quad (18)$$

Key properties:

- Over-filled epochs (more than one session in epoch i) incur no additional penalty.
- The minimum achievable penalty is 0, attained when every epoch contains at least one session — the ideal uniform distribution.
- The maximum possible penalty for a pair (b, c) is $\alpha \times R_{b,c}$ (all epochs empty), which in practice cannot occur because H1 requires exactly $R_{b,c}$ sessions to be placed.

Interaction with the Consecutive Rule

The spreading and consecutive rules are additive in the objective and fully independent in their constraint sets. In edge cases they may pull in opposite directions: spreading encourages a session in every period while the consecutive rule penalizes too many adjacent periods. The weight α resolves this tension — a small α lets the consecutive rule dominate, while a larger α prioritizes uniform distribution. The two rules may be enabled independently or together.